

Gokul Gopalan Ramachandran

PhD Candidate in Economics

Department of Economics, University of Warwick, Coventry CV4 7AL, United Kingdom

gokul.gr@warwick.ac.uk · gokulgopalan.github.io

EDUCATION

University of Warwick

2021 –

PhD in Economics

Supervisors: Eric Renault and Kenichi Nagasawa

University of Warwick

2021 – 2023

Master of Research in Economics

Distinction

Princeton University

2018 – 2020

Master in Public Affairs

GPA: 3.77

Economics and Public Policy

Field GPA: 3.92

National Institute of Technology, Calicut

2006 – 2010

Bachelor of Technology in Electronics and Communication

GPA: 8.53/10

FIELDS

Research Econometrics

Teaching Econometrics (PhD and undergraduate), Probability and Statistics (undergraduate)

JOB MARKET PAPER

Granular Instrumental Variables in Large Panels: Identification and Inference Across Strong, Nearly Weak, and Weak GIV [arXiv]

The methodology of Granular Instrumental Variables (GIV) pioneered by Gabaix and Koijen (2024) has become a popular way to estimate causal relationships involving aggregate variables in macro-finance. I study the estimation of such causal relationships in panels where both the cross section and time series are large ($N, T \rightarrow \infty$) using the GIV framework. I make two contributions. First, I show that the GIV estimators can consistently recover parameters for two different equations of interest and provide the correct asymptotic theory. Second, I show that GIVs can be weak in the absence of large units (lack of granularity), but the parameter of interest remains recoverable. The GIV estimator remains consistent and asymptotically normal, now at a slower rate. I characterise the asymptotic behaviour of this nearly weak instrument. I apply this theory to estimate the supply and demand elasticities in three markets: crude oil, copper, and natural gas. Copper and natural gas have dominant producers, yielding strong GIV. Crude oil has many small producers, leading to slightly weak GIV. All yield inelastic estimates.

WORKING PAPERS

Optimal Granular Instrumental Variables: Sequential Estimation of Long Panels with Finite Cross Section

with Kenichi Nagasawa and Eric Renault

We extend the theory of Granular Instrumental Variables for small panels (N fixed and $T \rightarrow \infty$) to address the question of optimal instruments and feasible estimation. Among the different ways of estimating the GIV, we characterise the choice that leads to the smallest variance. We also show that the first-step estimation of this feasible optimal instrument has an impact on the asymptotic distribution of estimators of the structural parameter. We fully characterise this distribution for two estimators: a GMM estimator and a PCA estimator.

Interactive Fixed Effects with Heterogeneous Parameters

with Kenichi Nagasawa

Bai's (2009) Interactive Fixed Effects estimator is a popular method to estimate parameters in panels with limited endogeneity, but the estimator fails when the parameter of interest varies with individual units. In this paper, we propose a new estimator that can handle individual heterogeneity using non-parametric estimation.

TEACHING

Senior Graduate Teaching Assistant

2022 –

University of Warwick

- EC9A3: Econometric Theory (PhD Year 1) *Spring 2024, 2025, 2026*
- EC140B: Mathematical Techniques B (undergraduate; Alex Dobson) *Fall 2024*
- EC203: Applied Econometrics (undergraduate; Tom Martin) *Fall 2023*
- EC124: Statistical Techniques B (undergraduate; Jeremy Smith) *Spring 2023*

RESEARCH EXPERIENCE

Research Assistant, Elizabeth Levy Paluck's Lab

Fall 2019 & Spring 2020

Professor of Psychology and Public Affairs, Princeton University

WORK EXPERIENCE

Between 2011 and 2018, and again in 2020–21, I was a civil servant in the career track of India's highest civil service: the Indian Administrative Service. I led major development initiatives for the State of Kerala, rising to be the Acting Secretary to the Government in the Budget Wing (third highest rank in the department). Some of the highlights of my work:

Acting Secretary to Government (Finance-Resources)

August 2020 – September 2021

Government of Kerala

- Chief of one of the two wings of the State Finance Department: Resources
- Fiscal planning and management: resource mobilisation of \$15 billion during the Covid pandemic

District Collector and Chief Executive Officer

August 2016 – July 2018

Idukki District

- Chief Executive of the Government in a district with a population of 1.1 million
- Designed, implemented and evaluated economic development projects

State Mission Director, National Health Mission

August 2015 – August 2016

Government of Kerala

- Designed and implemented health programs with a budget of \$100 million
- Part of the senior management team when Kerala reduced the Infant Mortality Rate from 12 to 6

ACADEMIC AWARDS

- Tutor Award for Outstanding Evaluation, University of Warwick
- Economic and Social Research Council DTP Award *2023 – 2027*
- Department of Economics Fellowship for MRes, University of Warwick *2021 – 2023*
- Karl E. Prickett Fellow in Public and International Affairs, Princeton University *2019 – 2020*
- Princeton University Graduate Fellowship *2018 – 2020*
- National rank of 19 (out of 650,000) in the All India Civil Services Examination *2010*
- Second in Regional Mathematical Olympiad *2004*

SERVICE

- PhD Representative, Student-Staff Liaison Committee, University of Warwick *2022 – 2025*
- Graduate Associate, Center for Public Finance and Policy, Princeton University *2019 – 2020*
- Elected member, Social Service Society, National Academy of Administration *2011 – 2013*

TRAINING

- Data Science Workshop, University of Warwick *Summer 2024*
Mateusz Stalinski, Nathan Canen, Rafael Jimenez
- Causal Machine Learning, Study Center Gerzensee *Summer 2024*
Victor Chernozhukov

SOFTWARE AND LANGUAGES

Software Matlab, Stata, R, Python, L^AT_EX
Languages English, Hindi, Malayalam

REFERENCES

- **Eric Renault**
 Department of Economics, University of Warwick, CV4 7AL, United Kingdom
 eric.renault@warwick.ac.uk
- **Kenichi Nagasawa**
 Department of Economics, University of Warwick, CV4 7AL, United Kingdom
 kenichi.nagasawa@warwick.ac.uk
- **Ao Wang**
 Department of Economics, University of Warwick, CV4 7AL, United Kingdom
 ao.wang@warwick.ac.uk